MILITARY SPECIFICATION

CHOLESTYRAMINE FOR ORAL SUSPENSION, MODIFIED

This specificaton is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Cholestyramine for Oral Suspension, Modified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, Pa. 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a powder containing four grams of anhydrous Cholestyramine Resin in suitable excipients, flavoring, and coloring agents to make nine grams of powder.

3.1.1 Assay. Shall assay to contain not less than 1.60 mEq chloride and not more than 2.30 mEq chloride per gram of powder when assayed as specified in 4.7.1.

3.1.2 Identification. The infrared absorption spectra shall exhibit maxima and minima only as that of Cholestyramine Resin reference standard when determined as specified in 4.7.2.

3.1.3 Loss on drying. Loss on drying shall not be more than 8.50 percent when 3.0 gram sample of the powder is dried to constant weight by heating in a vacuum oven at 100° C for 4 hours (maximum mercury pressure 50 mm).

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3.2 Ingredients.

3.2.1 Cholestyramine Resin.

3.2.1.1 Assay. Shall assay to not less than 3.7 mEq chloride and not more than 4.8 mEq chloride per gram (on an anhydrous basis) when assayed as specified in 4.7.3.

3.2.1.2 Identification. The infrared absorption spectra shall exhibit maxima and minima only as that of Cholestyramine Resin reference standard when determined as follows:

Weigh 100 mg of sample and place in the grinding vial with steel balls; Add 0.50 gram Potassium Bromide and grind for one minute. Continue as specified in 4.7.2 beginning with "Using a suitable die, fill with the ground mixture..."

3.2.1.3 Loss on drying. Loss on drying shall not be more than 12.0 percent when a sample of 3.0 grams is dried to constant weight by heating in a vacuum oven at 70° C for 16 hours (maximum mercury pressure 50 mm).

3.2.1.4 pH. When 1.0 gram of sample is suspended in 100 ml of purified water, the pH shall be not less than 4.00 and not more than 6.00 at 25° C, when determined potentiometrically, using the USP method.

3.2.1.5 Residue on ignition. Shall have a residue on ignition of not more than 0.10 percent on an anhydrous basis when determined in accordance with the USP residue on ignition method.

3.2.1.6 Heavy metals. Shall have a heavy metals content of not more than 0.002 percent, on an anhydrous basis, when determined in accordance with USP, Heavy Metals, method II procedure.

3.2.1.7 Dialyzable quaternary amines. Shall contain not more than 0.05 percent of dialyzable quaternary amines when determined by an appropriate, accurate, and reproducible method. The bidder or offeror shall submit with its bid or proposal the proposed method used to determine the dialyzable quaternary amines. If the method is approved and accepted by the Government, the method shall be employed in the testing of the material. The requirement for submission of the above method may be waived, provided the offeror states, in his bid or proposal, that the product he proposes to furnish is the same product he has offered to the purchasing activity on a previous procurement and that the suppliers of the ingredients, specification of the ingredients, and the method of manufacture of the finished product have not been changed.

3.3 Other ingredients. All other ingredients entering into the manufacture of the finished product shall be of USP or NF quality or, if not included in either of these compendia, the ingredients shall be suitable for use in the product.

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3.4 Net contents. The net contents (fill weight) of each individual packet shall be not less than 7.73 grams and not more than 10.46 grams.

3.5 Expiration dating period. The expiration dating period shall be 48 months.

3.5.1 Delivery. Not more than 4 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

3.6 Workmanship. The material and its containers shall be free from any defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of product produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

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4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Unit of issue

4.3.1 Raw material and end item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Criteria for raw material and end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Assay	3.1.1	S-2	5 } Composite
Identification	3.1.2	S-2	5 } Composite
Loss on drying	3.1.3	S-2	5 } Composite
Net contents	3.4	S-4 (AQL-0.40)	32 (individual immediate container)
Ingredients: 2/			
Cholestyramine Resin			
Assay	3.2.1.1	-	-
Identification	3.2.1.2	-	-
Loss on drying	3.2.1.3	-	-
pH	3.2.1.4	-	-
Residue on ignition	3.2.1.5	-	-
Heavy metals	3.2.1.6	-	-
Dialyzable quaternary amines	3.2.1.7	-	-

1/ Where possible, the sample shall be used for two or more tests.
2/ Sufficient quantities of each of the raw materials shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

4.4 Examination. The finished product shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3.

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4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects.

Categories	Defects 1/
Major:	
101	Finished product not free from foreign matter.
102	Finished product not homogeneous.
Minor:	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks or tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests.

4.7.1 Assay (finished product).

Reagents.

Silver Nitrate, 0.1N Standard Solution.
Nitric Acid, Concentrated, 70 percent.
Nitrobenzene, Analytical Reagent.
Ferric Ammonium Sulfate, 8 percent W/V Solution.
Ammonium Thiocyanate, 0.1N Standard Solution.

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Procedure.

Transfer about 2.0 grams of sample, accurately weighed, to a 300-ml glass-stoppered Erlenmeyer flask and add exactly 50.0 ml of 0.1N Silver Nitrate. Add 15 ml of Nitric Acid and boil until dissolved (about 10 minutes).

Boil until solution becomes clear.

Filter the hot solution through fluted filter paper. Wash residue with four 10-ml portions of hot purified water. Cool and add purified water (cool) to make about 150 ml.

Add 5 ml of Ferric Ammonium Sulfate indicator solution and titrate with -.1N Ammonium Thiocyanate to a permanent faint brown color. (Nitrobenzene addition may be necessary to enhance the titration endpoint.)

Calculation:

$$\frac{(\text{ml Silver Nitrate} \times N) - (\text{ml Ammonium Thiocyanate} \times N)}{\text{Grams of sample}} = \text{mEq Chloride per gram}$$

Sample may be titrated on an autotitrator.

4.7.2 Identification (finished product).

Place 5.0 grams of sample in a 100-ml beaker. Add 50 ml of 0.1N Hydrochloric Acid solution and stir to suspend the sample; heat on a steam bath for about 10 minutes. Filter the hot suspension through Whatman No. 54 filter paper; wash the material on the filter with three 20-ml portions of purified water and discard the filtrate and wash water.

Dry the washed material in an air oven at 100° C for 2 hours and cool to room temperature. Weigh (within 0.2 mg) 0.0025 gram of the dried substance and transfer to a suitable grinding vial with steel balls; add 0.50 gram Potassium Bromide and grind for one minute.

Using a suitable die, fill with the ground mixture (sample and KBr) to a depth of 0.031 inch. (Use the base of die to hand press level full and to remove excess of mixture from die.) Place base in position, invert die assembly, place on base, and evacuate in a suitable evacuating bell for 2 minutes. Continue evacuation while pressing (with a suitable press) at 3,500 - 5,000 psi for 3 minutes. Transfer prepared pellet to holder and record absorption from 2 to 16 microns using air as a reference. Compare spectrum with Cholestyramine Reference Standard.

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4.7.3 Assay (ingredient).

Reagents.

Silver Nitrate, 0.1N Standard Solution.
Nitric Acid, Concentrated, 70 percent.
Nitrobenzene, Analytical Reagent.
Ferric Ammonium Sulfate, 8 percent W/V Solution.
Ammonium Thiocyanate, 0.1N Standard Solution.

Procedure.

Weigh accurately approximately 0.5 gram of sample and transfer to a 250 ml, glass-stoppered flask. Add 50 ml of purified water and swirl to dissolve sample. To each flask, add exactly 50.0 ml of 0.1N Silver Nitrate, 3 ml of Nitric Acid, 5 ml of Nitrobenzene, 2 ml of Ferric Ammonium Sulfate Solution and shake well. Titrate the excess of Silver Nitrate with 0.1N Ammonium Thiocyanate to a permanent faint red-brown color. Nitrobenzene may be used to enhance the endpoint.

Calculation:

$$\frac{(\text{ml Silver Nitrate} \times N) - (\text{ml Ammonium Thiocyanate} \times N)}{\text{gram of sample}} \times \frac{100}{(100\text{-percent L.O.D. } 1/)} = \text{mEq chloride per gram, anhydrous basis}$$

Sample may be titrated on an autotitrator.

1/ L.O.D. = Loss on Drying.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1),

5.2.1 Unit of issue. One box containing fifty (50) immediate containers, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

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5.2.2.2 Unit packaging. Each unit shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate packaging. Six (6) units of issue shall be packaged as specified in 5.2.8 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purposes.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Twenty-four (24) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twenty-four (24) units of issue shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.
(c) the expiration date.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.

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5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. In addition, Type I shelf life marking as specified in MIL-STD-129 shall be shown.

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-00-105-0372	CHOLESTYRAMINE FOR ORAL SUSPENSION, MODIFIED, Equivalent to 4 Grams of Anhydrous Cholestyramine, 50s

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA - DM

Project No. 6505-1693

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-37536 CHOLESTRYAMINE FOR ORAL SUSPENSION, MODIFIED

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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